Reproducibility study of "Explaining RL Decisions with Trajectories"

Abstract

Reinforcement learning (RL) models are becoming increasingly more complex and capable and are being used for more critical applications, stimulating the need for explainable RL (XRL) in the wider field of explainable AI. We study the reproducibility of the XRL work by Deshmukh et al. (2023), who present a method using training data attribution of trajectory clusters to explain the action of offline RL agents. Our study examines the authors' claims that tokenization and embedding of trajectories can yield semantically meaningful clusters, and that humans can interpret these clusters to understand an agent's decisions. Our replication, despite successfully reproducing several of the original paper's technical steps and achieving similar quantitative results, indicates that the produced clusters lack a shared semantic meaning, contrary to the original findings. This highlights challenges in reproducing some aspects around the human evaluation of explanation interpretability. We try to determine the sources of the inconsistencies in our findings, amend them, and furthermore, try to extend the method. Our results underscore the need for further research to address these challenges and explore more effective strategies for explainable reinforcement learning.

1 Introduction

Reinforcement learning (RL) has been successfully utilized across various fields, including robotics, gaming and finance. A persisting challenge is that RL algorithms depend on interacting with a live (online) environment for learning, complicating their application in scenarios where direct engagement with the environment is expensive, impractical, or dangerous. This approach has proven especially successful in areas such as finance, healthcare and autonomous driving Levine et al. (2020). However, as RL models become more complex due to the use of deep learning and their applications more critical, the need for explainability of these models becomes pressing.

Several approaches in explainable RL (XRL) have been identified like approximating more complex models with simpler ones (Li et al., 2021) or saliency-based methods Heuillet et al. (2021). The method proposed by Deshmukh et al. (2023) is to explain a decision of an offline-trained RL agent by pointing to a training experience (trajectory) that most influenced it to perform this action. In this work we aim to evaluate this method by attempting to replicate some of the results in Deshmukh et al. (2023) and by evaluating the individual steps of the process of finding the explanations.

1.1 Structure of the report

In the Scope of Reproducibility section, we further elaborate on the work of the original authors and examine the claims made in the original paper. In the Methodology section, we explore datasets and environments on which we apply the proposed method. We also propose experiments that test the claims quantitatively and human evaluation experiments that verify the interpretability of the proposed method. We also discuss computational requirements and our novel ideas for improving the method. In the Results section, we present and analyze the results of our experiments. We describe the number of participants in the human evaluation study and observed issues. The discussion section discusses the results and challenges that occurred during the reproduction of the results, summarizing which claims hold and which may not generalize well, as well as possible suggestions for taking the work further.

2 Scope of reproducibility

The original paper contributes an attempt at increasing the explainability of reinforcement learning algorithms. The approach that the authors take hinges on that a decision of a trained agent can be explained by pointing to an example encountered during training which caused the agent to make its decision. The naive method of finding such an example would be to remove trajectories one at a time and retrain the agent on the remaining training data, until a training example without which the agent would make a different decision in the state being considered is found. This approach is computationally infeasible. The work by Deshmukh et al. (2023) is the first to attempt to provide a tractable method for finding responsible trajectories that are suitable as human-understandable explanations.

2.1 Overview of the method

The method proposed by the authors is composed of several steps:

1. First the raw training trajectories are tokenized and passed through a sequence encoder (LSTM or Transformer) for the purpose of embedding them in a semantic space.
2. Next, the trajectories are clustered, with the intention of obtaining semantically meaningful clusters of similar trajectories.
3. Each cluster of trajectories is encoded into a single data embedding, for the purpose of measuring distance between them.
4. For each cluster, an explanation agent is trained on a subset of the full data, omitting the respective cluster. An additional agent is also trained on the full dataset.
5. The top-k policies with the most different actions relative to the fully trained agent according to some action distance are selected.
6. Finally, the respective top-k data embeddings of the subsets are compared using to the to the embedding of the full dataset using Wasserstein distance and the most similar one is selected. The omitted cluster of that data subset serves as the explanation of the agent's action.

2.2 Claims of the original paper

The paper's proposed method aims to produce meaningful, human-understandable explanations for actions proposed by an RL agent to a given state. This main claim can be further divided into more specific ones with accompanying evidence which we found to be as follows:

1. A combination of tokenization and embedding methods inspired by NLP methods can capture semantically meaningful information about the training trajectories. This claim is supported by indirectly by visual inspection of the embedded data after PCA (see Appendix A4 and A8 of Deshmukh et al. (2023)).
2. Clustering the trajectories in the embedding space produces distinct clusters with their own clear meanings that are interpretable to humans. This claim is supported through visual inspection of the clusters and post-hoc attribution of meaning to them (see Section 4.2 and Appendix A4 and A8 of Deshmukh et al. (2023)).
3. Explanations of an agent's behaviour are meaningful and informative to humans. This is supported by a human study, where they are compared to random explanations (see Section 4.3 of Deshmukh et al. (2023)).

3 Methodology

The method proposed by the authors is originally implemented in four environments: GridWorld, SeaQuest, HalfCheetah, and Breakout. Out of these four, we reproduced the whole method for GridWorld and the

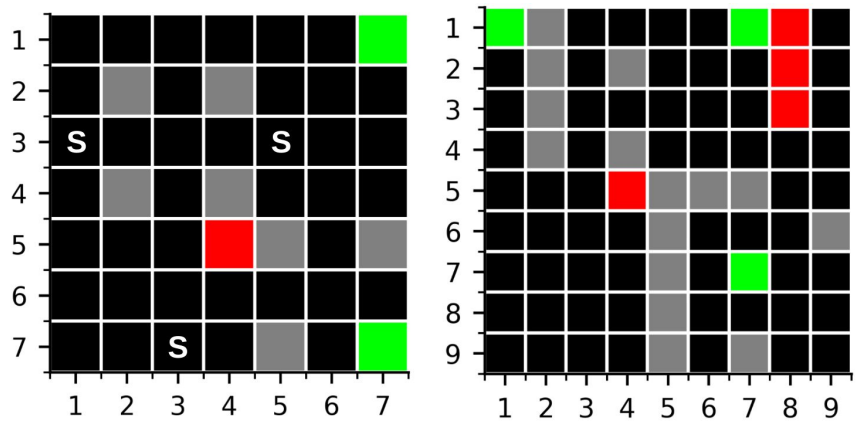


Figure 1: Original 7x7 Grid used by the Authors, with start points denoted by letter "S". On the right is our larger 9x9 Grid, where every non-terminal, non-obstacle cell is a starting point. The red terminals are "lava" and have a large negative reward. The green terminals are the goals and have a large positive reward. The grey cells are obstacles.

embedding and clustering part for Seaquest. The code for the GridWorld implementation was made available to us by the authors upon request. We ran this code without modification and with modifications to the grid size, data generation, and clustering approach. The code for Seaquest was created manually, using existing datasets and a pre-trained sequence encoder described in the original paper.

3.1 Environments and datasets

3.1.1 Grid World

GridWorld consists of a simple, 2D grid-based environment with green goal cells, red lava cells and gray obstacles, where the agent's role is to reach the goals cells and avoid lava cells. We generate the offline data for GridWorld using partially trained Dyna-Q Agents further described below. For the default 7x7 grid, we generate 50 trajectories ending in a green terminal and 10 that end in a red terminal. The max length of the trajectories is 10, which is the exact setup used by the authors. It is worth noting that the original 7x7 Grid has only three starting points - this was apparent in the code that we received from the authors.

For our custom, larger 9x9 Grid (Figure 1), we set starting points in all free cells (that are not terminals or obstacles). This way, in a 9x9 Grid, there are 59 starting points, which is almost exactly 20 times more than in a 7x7 Grid. We also generate 20 times more offline trajectories: 1200 in total (1000 ending in the green terminal, 200 in the red terminal). We kept the original max length of 10.

3.1.2 Seaquest

For SeaQuest, we use an existing 'seaquest-medium-v0' dataset from the Google Dopamine datasets Castro et al. (2018), accessed through the d4rl-atari library [1] - an adaption of d4rl library Fu et al. (2020) for Atari games. This is the same dataset the authors used, with 717 trajectories and 1000000 observations. This dataset is split into sub-trajectories of length 30. For our needs of assessing the clusters, we only used a 1000 random subtrajectories.

3.2 Model descriptions

GridWorld The data generation agents were trained using the Dyna-Q algorithm. The policy agents were trained using dynamic programming on the probabilistic transition and reward model as estimated from their respective data subsets. We used the same implementation as the original paper's authors for both

[1] https://github.com/takuseno/d4rl-atari

agents. For the encoder, we use the encoder part of an LSTM-based Seq2Seq Encoder-Decoder model. For the hyperparameters of the agents and encoder models, see Appendix A.

SeaQuest To generate embeddings for SeaQuest trajectories, we used a pre-trained decision transformer Chen et al. (2021), used by the authors of the original work.

3.3 Experimental setup and code

Here we list the experiments we ran to reproduce and extend the results of the original paper. The description of the experiment is accompanied by whether it test a a claim as seen in Section 2.2 or whether it is an extension.

3.3.1 Human understandability study

We ran a human understandability study similar to the one presented in the original paper to evaluate the performance of the method in finding explanatory trajectories. We conducted a survey, in which respondents were presented with 7 example states and actions of a fully-trained RL agent. For each state and action, respondents were shown four possible explanation trajectories and asked to choose the two that they thought explained the action best. Out of the two trajectories from the responsible cluster, one was picked at random, and one was picked by finding a trajectory that contained a matching state and action (if possible, otherwise random), as was done by the authors of the original paper. In addition, two random trajectories were presented, where one was ensured that it is from outside the responsible cluster, while the other was simply selected at random. With this setup we aimed to replicate the levels of human preference of explanations found by the method as compared to presenting a random training trajectory. We also wished to evaluate to what extent the sampling method (matching state and action) was responsible for the human preference.

3.3.2 SeaQuest: Human Semantic Assignment

In the second study, which consisted of a pilot project and a main study (Figure 2), we captured how humans semantically understand and group SeaQuest trajectories. Our objective is to compare clusters organically formed by humans and intended to bear semantic meaning with those identified by the original research methods. Our participants developed categories like 'Oxygen management', which align with the semantic cluster meanings the authors describe. Hence, we expect our clusters to match the clusters found by the algorithm. Our study, therefore, tests claim 2 (embeddings produce meaningful clusters). Throughout this study, trajectories were presented as 30-frame animated GIFs that showed the agent's interactions with the SeaQuest environment.

During the pilot, the participants freely developed short, meaningful labels (maximum eight words) that described individual trajectories. Next, we consolidated these 120 labels into eight broader semantic categories to potentially find support for the same number of clusters the original method uses. For example, we grouped labels like 'going up', 'ship going to the right' and 'going deeper' into the category 'Movement and Navigation' Appendix C.

In the second part of the study, new participants assigned the most appropriate category out of the eight categories developed in part one, and optionally the second-best fitting category, to new individual trajectories. These new trajectories were 32 total trajectories - four randomly sampled trajectories per cluster that has been found by the original algorithm. To compare our human-developed clusters to the original clusters, we checked if participants assigned trajectories of the same cluster with homogeneous labels (claim 2). We further tested if the human-developed clusters were distinct in feature space and how many original clusters they were composed of.

3.3.3 Embedding and cluster analysis

Here, we quantitatively analyze the embeddings and test whether they encode meaningful characteristics of the trajectories. Specifically, we correlate each embedding dimensions with the length, total reward, whether the trajectory reaches a goal, and the average direction of the trajectory in radians as given by the vector

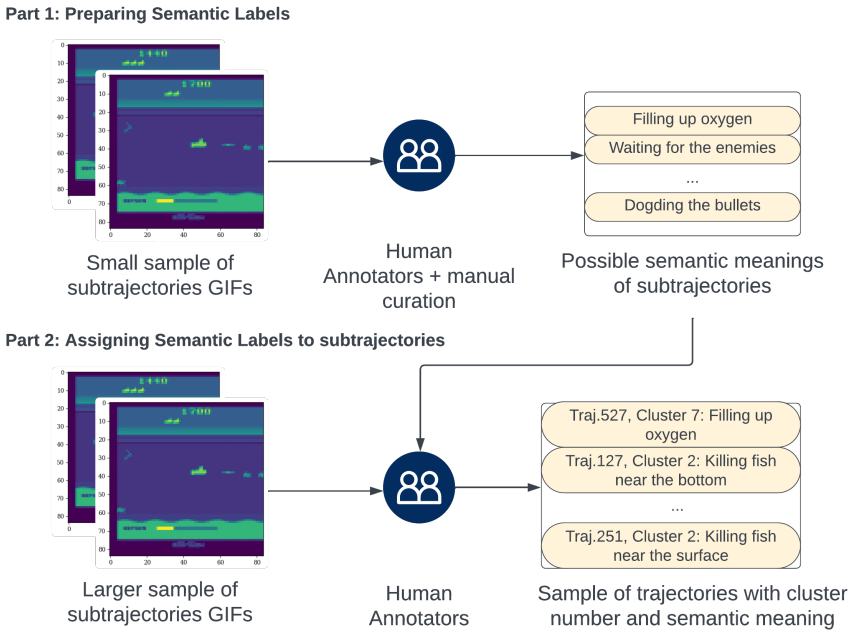


Figure 2: Overview of the human semantic assignment study to understand the semantic meaning of the clusters. Note that the labels used here do not correspond to true labels, which you can find in Appendix C.

subtracting the start from the end position. We do this for the original 7x7 GridWorld. We then calculate the average absolute correlation and R^2 averaged embedding dimensions and their standard deviation.

We further tested the embeddings acquired when using an increased amount of epochs ($25 \rightarrow 100$), using more trajectories ($60 \rightarrow 300$), and filtering duplicate trajectories. This tests Claim 1 and Claim 2 of the original paper.

3.3.4 Recursive clustering

The authors of the original paper want to "find the smallest set of trajectories, the absence of which from the training data leads to different behavior at the state under consideration" (p. 3, Deshmukh et al. (2023)) while keeping the computational feasibility. In this extension of the original paper, we want to answer the question: if not for computational feasibility, how small sets of responsible trajectories could be? An answer to this question may help direct further research and show the potential of explaining the model's decision with a small set of responsible trajectories.

We suggest a method that allows us to find a very small set of trajectories, the absence of which leads to different behavior of the original agent in a particular state.

Suggested Method Instead of performing one initial data clustering, we suggest recursive clustering. When asked to explain a given state and action, recursive clustering will first split the data into k clusters with the KMeans algorithm and check which cluster absence changes the prediction like in the original method. In the next step, we cluster only data in the previously selected cluster and choose the responsible subcluster. By repeating this step, we iteratively narrow the number of responsible trajectories, training at most $O\left(k \log_k \left(\frac{n}{k}\right)\right)$ models to get at most k responsible trajectories (Assuming that n is the number of all trajectories). For an overview of the whole algorithm, see Appendix D.

We compare the number of found explanations and size of responsible trajectory sets of recursive clustering vs. the original method on our 9x9 Grid. Here, we set the $k = 10$, which requires training around 20-30 models to explain one particular model action in a given state.

Table 1: Reproduction of Quantitative Analysis of Grid-world Trajectory Attribution. The analysis is provided using five metrics, consecutively: Initial State Value Estimate, Local Mean Absolute Action-Value Difference, Action Contrast Measure, Distance between embeddings, Cluster Attribution Frequency.

π	$\mathbb{E}(V(s_0))$	$\mathbb{E}(\Delta Q_{\text{orig}})$	$\mathbb{E}(1_{(\tau_{\text{orig}}(s)\neq\tau_j(s))})$	$W_{\text{dist}}(d_i, d_j)$	$P(\text{final} = c_j)$
orig	**0.3061**	-	-	-	-
0	0.3055	0.0157	0.0204	0.00018	0.0000
1	0.3053	0.0251	0.0408	0.0005	0.0000
2	0.3049	0.0202	0.1224	0.00017	0.0000
3	0.2989	0.0144	0.0408	0.00017	0.1250
4	0.3054	0.0302	0.1224	0.00003	**0.5000**
5	0.3057	0.0014	0.0204	0.00015	0.0000
6	0.3046	0.0291	0.1224	0.00021	0.0000
7	0.2857	**0.0600**	0.0816	0.00069	0.3750
8	0.3057	0.0115	0.0204	0.00014	0.0000
9	0.3046	0.0234	**0.1429**	**1.0000**	0.0000

3.4 Computational requirements and environmental impact

For GridWorld experiments, (both reproducing the original results, and the recursive clustering method) we spent around 10 hours of CPU time (Consumer PC, 16GB RAM, i5-8th gen). We spent 2 hours of CPU time for SeaQuest trajectory encoding and clustering. We estimate the electricity usage needed as less than 1kWh, which results in around 0.5kg CO2 emissions in the Netherlands.

4 Results

4.1 Reproduction of original paper results

4.1.1 Reproduction of the RL agents

We trained GridWorld agents on different subsets of the embedded data, and evaluated their performance using the same metrics as the original authors. We also show distances between embeddings of original and complementary data sets with Wasserstein metric (normalized to [0, 1]), and the cluster attribution frequency. The results are in Table 1

We see that the results do not exactly match the results obtained by the authors, but the highest value matches in 4 out of 5 metrics and the magnitude is similar for all metrics, except the distance, where reproduced distances are lower than ones obtained by the original authors. This may be caused by the stochasticity in the embedding model training. We conclude that the agents for GridWorld are reproducible.

4.1.2 Human understandability study

In the GridWorld explanation evaluation survey, participants chose explanations from within the responsibility cluster 59.9% of the time, which is considerably lower than the 70% reported by the authors of the original paper. We also observed that it was the state/action matched explanations that the respondents preferred, while random explanations from the responsible cluster failed to outperform random explanations from other clusters.

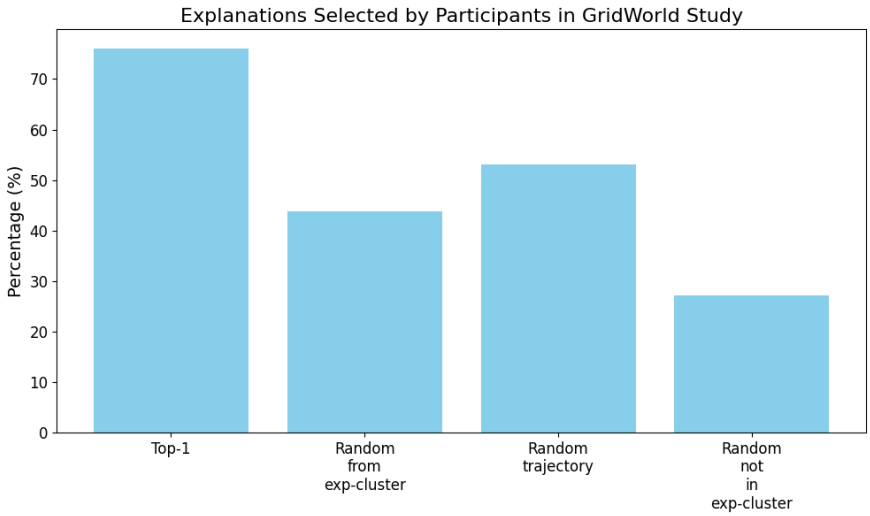


Figure 3: Results of survey assessing the quality of explanations in GridWorld

4.1.3 Human semantic assignment study

Firstly, we found that our participants assigned different labels to trajectories of the same cluster, as shown in Figure 4. Parallely, the participants labelled individual trajectories consistently, reflected in a lower average trajectory entropy (all < 1.3) than cluster entropy for every cluster (all > 1.9) (Appendix E). Secondly, when grouping the trajectories by the label the participants assigned to them, we observe no pattern in PCA feature space (Figure 5), with the human-developed clusters even having a higher total within-cluster error (WCE = 78.84) than the original clusters (WCE = 54.67). Further, the human-developed, meaningful clusters did not map to the original clusters: one the one hand, four out of the eight labels, and hence, four clusters, were prioritized. On the other hand, those clusters included a heterogeneous mixture of sample trajectories, that stemmed from at minimum five and up to eight different original clusters.

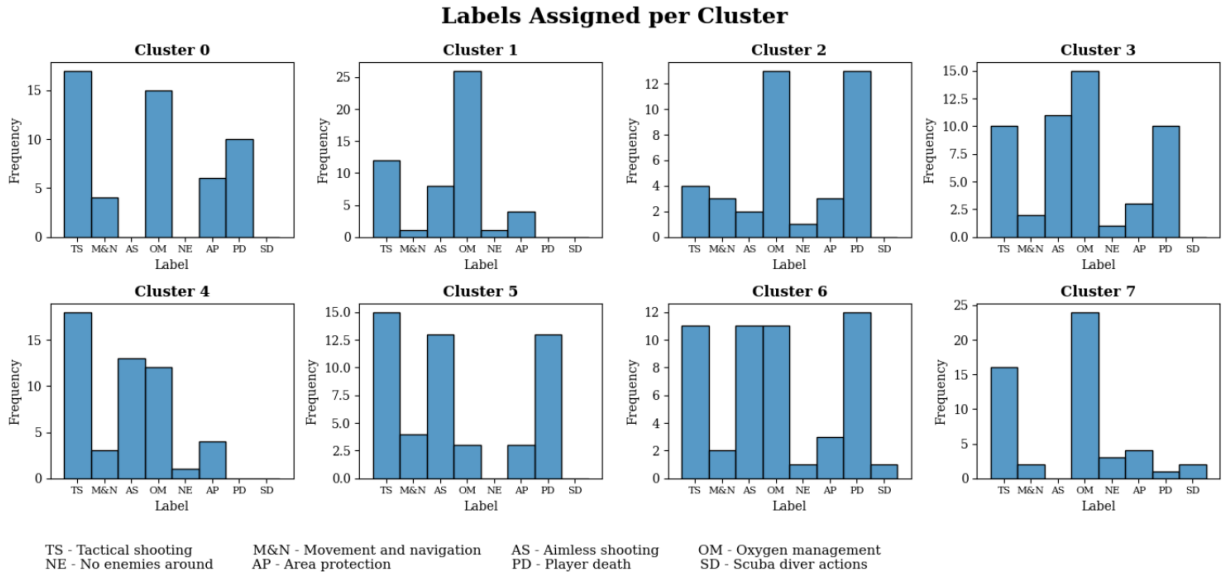


Figure 4: Distribution of labels assigned to each cluster's trajectories.

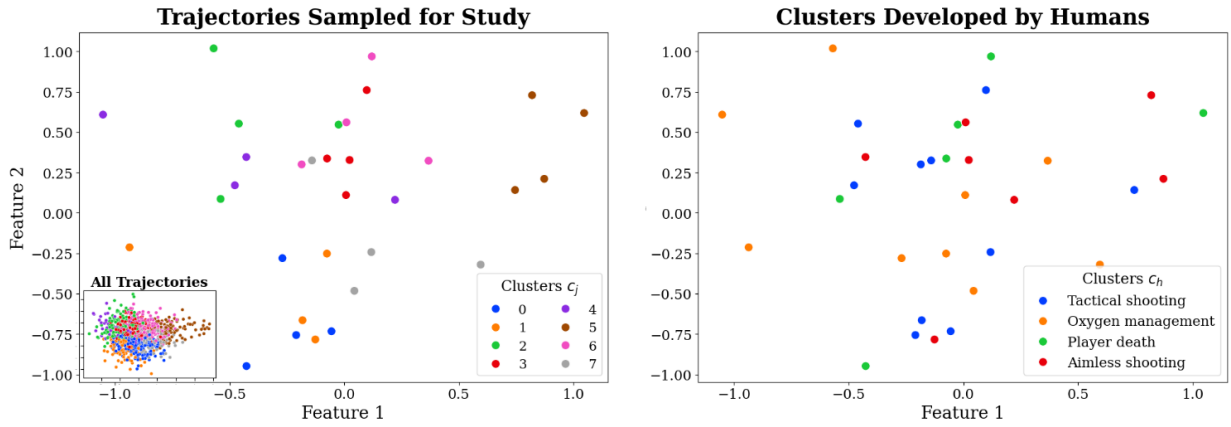


Figure 5: Comparison of the trajectory sample's original clusters with the human-developed, meaningful clusters

These findings reject that the reproduced method fulfills claim 2, as none of the original clusters shared a meaning according to our participants, and the meaningful human-developed clusters did not distinctly manifest in feature space.

4.1.4 Embedding and cluster analysis

Table 2: Correlation between embedding dimensions and trajectory characteristics

Attribute	$\mathbf{r^2_{original}}$	$\mathbf{r^2_{traj+filter}}$	$\mathbf{r^2_{epoch}}$	$\mathbf{r^2_{traj}}$	$\mathbf{r^2_{traj+epoch+filter}}$	$\mathbf{r^2_{traj+epoch}}$
Length	0.59 ± 0.20	0.51 ± 0.25	0.36 ± 0.28	0.37 ± 0.33	0.23 ± 0.30	0.22 ± 0.27
Angle	0.07 ± 0.12	0.12 ± 0.22	0.16 ± 0.17	0.31 ± 0.35	0.34 ± 0.32	0.35 ± 0.27
Reward	0.09 ± 0.15	0.05 ± 0.08	0.28 ± 0.33	0.17 ± 0.17	0.16 ± 0.17	0.20 ± 0.21
Goal %	0.07 ± 0.12	0.14 ± 0.20	0.28 ± 0.34	0.28 ± 0.36	0.36 ± 0.31	0.35 ± 0.27

The results of the embedding analysis can be seen in Table 2. The r^2 values for the original configuration largely encode length instead of the other attributes. Increasing the number of encoder training epochs or trajectories seems to yield more balanced embeddings that encode different aspects of the trajectories, and combining them further shifts emphasis to attributes other than length. Filtering duplicates does not seem to have a sizable effect on the embeddings other than reducing the number of training trajectories as the original and added filtered trajectories embeddings seems to behave similarly.

4.2 Extensions of original paper and novel results

4.2.1 Recursive clustering

We used the Recursive Clustering method, as well as the original authors' method on the 9x9 GridWorld environment with 1200 offline trajectories.

The original method found the explanation for the agent behavior in 18 states, and recursive clustering found the explanation in 14 states, out of which 13 are small (number of trajectories fewer than k=10). Twelve states/actions were explained both by recursive clustering and the original method - the number of explanations for commonly explained states is summarized in Table 3.

We observe the true number of responsible trajectories is very small (2-10) for many states/actions. This may suggest that RL Agents' behaviors can be explained in reality by very few trajectories, making the

Table 3: Number of found responsible trajectories per state, which absence change the model's behavior. The number of trajectories for both original method and recursive clustering. The original set of responsible trajectories does not always contain the new one because of the different clustering method.

State	20	52	45	54	55	57	63	59	29	64	32	14
#Traj. Original	51	40	51	51	51	51	51	47	45	56	51	51
#Traj. Recursive	2	2	5	5	5	5	5	6	7	8	10	48
Recursive $\subseteq$ Original	F	F	T	T	T	T	T	T	T	T	F	F

explanations way more interpretable. This result may guide and encourage further research in quickly finding these small sets of responsible trajectories, and the proposed recursive clustering algorithm may help identify the smallest possible number of trajectories responsible for one particular state/action.

5 Discussion

We attempted to reproduce and extend the work by Deshmukh et al. in explaining actions of RL agents using attribution of training trajectory clusters.

5.1 Assessment of reproducibility

5.1.1 Challenges

- Lack of code for environments other than Grid-world. It made it more difficult to reproduce the results authors obtained in the other environments. Reproducing SeaQuest clustering required further communication with the authors.
- Vague descriptions of some steps in the methodology, eg. how trajectories were sampled from the responsible cluster. This made it difficult to precisely reproduce the human study.
- Finding a pre-trained transformer is not possible for all environments, so a new decision transformer would have to be trained for most other environments. This made it inconceivable for us to validate the method in a more realistic environment.

5.2 Communication with original authors

We received the codebase for GridWorld from the authors upon request. We further communicated with the authors to receive the details the dataset used in the SeaQuest Environment. Authors responded to questions in around 2 days.

5.3 Conclusions

We conclude that the paper "Explaining RL Decisions With Trajectories" is partially reproducible: each particular step of the method can be reproduced with similar quantitative results as the authors. On the other hand, human evaluation results contradicts authors' claims of interpretable and semantically meaningful clusters. Furthermore, we identify limitations of the method, like not satisfying embeddings of trajectories and challenges in regard of training a sequence to sequence model.

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A Hyperparameters of models

A.1 Grid-World

Agents The agent was trained using DynaQ - Dynamic programming for finding optimal policy values based on a reward model. We used $\gamma = 0.95$ and threshold $= 10^{-4}$.

Embedding Model For the encoder, we use the encoder part of an LSTM autoencoder with a token embedder of size 16 and a hidden dimensionality of 32. The embeddings obtained by the encoder are the concatenated hidden and context vectors of size 64. It was trained using stochastic gradient descent with 25 epochs at a learning rate of $\eta = 1$. This is the original configuration as in the original paper. We will change some of these values in the upcoming experiments.

B PCA plots of trajectory clusters

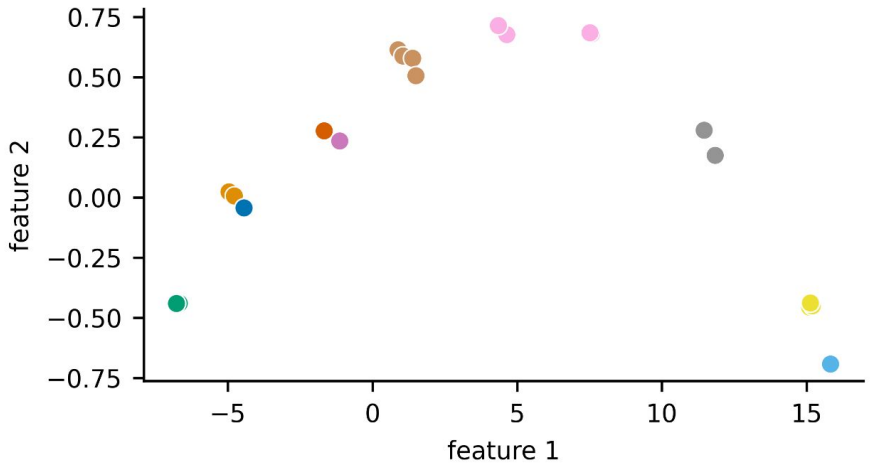


Figure 6: Reproduced clusters after PCA for the 7x7 Default Grid in GridWorld

C Human-derived Trajectory Labels and Groups

The eight main semantic categories which resulted from the pilot study and a selection of the 120 labels they are based on.

C.1 Tactical Shooting

- shooting enemies
- ship killing enemies
- ship shooting enemies to the right
- shooting one fish
- shooting enemies on the left of the screen
- hitting one fish
- ship shooting enemies
- Tactical movement and shot depending on fish

C.2 Movement and Navigation

- diving deep away from fish
- going up
- ship going to the right
- going deeper
- ship low behaving strangely
- ship moving in both direction and shooting

C.3 Aimless Shooting

- Aimlessly shooting to protect surface layers
- shooting and shooting
- ship missing enemies and shooting one
- Shooting in opposite direction of enemy
- ship shooting randomly
- hitting a fish with one of three shots

C.4 Oxygen Management

- Replenishing oxygen
- Low on oxygen
- Running low on oxygen
- filling up oxygen
- Awaiting Refilling oxygen
- running out of oxygen

C.5 No Enemies Around

- no fish
- no enemies in sight
- ship is resting
- waiting for the enemies

C.6 Area Protection

- shooting near the surface
- Defending a safe position
- Near miss while hovering in the top right corner
- Aimlessly shooting to protect safe position (top-right)

C.7 Player Dying

- starting a new round after losing one life
- Dying from lack of oxygen

C.8 Scuba Diver Actions

- killing the scuba diver
- catching the scuba diver
- collecting a scuba diver

D Recursive clustering algorithm

Algorithm 1 RecursiveClusteringExplainStateAction

not_responsible_trajectories ← {} ▷ Initialize the sets of trajectories
candidate_trajectories ← all_trajectories
$a_{\text{orig}} \leftarrow \pi_{\text{orig}}(s)$ ▷ Original action in given state
while length(candidate_trajectories) $> k$ **do**
    $K \leftarrow$ KMeans(candidate_trajectories) ▷ Run clustering on the candidates
    $C \leftarrow \{\}$ ▷ Set of clusters that explain original action
    for d_k in K **do**
        $d_{\text{temp}} \leftarrow$ not_responsible_trajectories $\cup$ (candidate_trajectories $\setminus$ d_k)
        $\pi_k \leftarrow$ train_agent(d_{temp})
        $a_k \leftarrow \pi_k(s)$
        if $a_k \neq a_{\text{orig}}$ **then** ▷ If agent behaves differently, get Wasserstein distance
            $C \leftarrow C \cup \{(d_k, W_{\text{dist}}(\text{all_trajectories}, d_{\text{temp}}))\}$
        end if
    end for
    if length(C) $= 0$ **then return** candidate_trajectories ▷ Early stopping
    end if
    responsible_cluster ← argmin(C) ▷ Choose cluster with min data embedding dist
    not_responsible_trajectories $\cup =$ candidate_trajectories $\setminus$ responsible_cluster
    candidate_trajectories ← responsible_cluster
end while return candidate_trajectories

E Human semantic assignment study: entropy

Cluster	Cluster Entropy	Average Single Trajectory Entropy
0	2.146223	0.851802
1	1.907536	0.740786
2	2.049922	0.647733
3	2.434083	1.169869
4	2.163821	1.033660
5	2.293512	1.210345
6	2.547830	1.124592
7	2.031325	0.978843

Table 4: Cluster Versus Average Trajectory Entropy. Measures dispersion of label assignment.